UNIVERSITY OF HYDERABAD
School of Physics
CENTRAL UNIVERSITY P.O.

Tel: +91-**(040)23134330**
Fax: +91-40-23010227
e-mail: **srinath@uohyd.ac.in**

Email ammanabrolu@uohyd.ac.in

8331879303

HYDERABAD 500 046

INDIA

Ref No: UoH/DST FIST/SS/ARK/01

CPP portal Tender ID: 2025_UoH_795091_1. Link: ePublishing System, Government of India

Tender notice supply of Multitarget UHV sputtering system:

** Please notice the Pre-Qualification criteria for the bidders in page no 3.

1. Multi target UHV Sputtering System
2. The system should consist of deposition chamber and load lock chamber with Ultra high vacuum.
3. The sputtering chamber should be provided with 6 sputter cathodes of DC.
4. Load lock chamber with a rf sputtering gun for deposition of oxide materials.

   Specifications

   4-1. Sputtering chamber specs

   Material with SUS304 stainless steel only.

   With a dimension of ϕ 400 × 600

   ~ Ultimate pressure of 9 x 10^{-8} Pa.

   The chamber should have many kinds of flange ports as follows

   6 ports of r Sputter cathodes

   1 Port for Vacuum pump (Turbo molecular pump with throttle butterfly valve)

   1 Port for Vacuum gauge port of UHV

   1 port for substrate holder with heating system

   1 Viewing port

5. Substrate specifications

Substrates size 2 inch diameter
Substrate is able to heat up to 750° C or better
Cathode shutters with manual operation
Target to substrate distance is about 150 mm (Distance should be optimized with the magnetron design)
Substrate heating with a temperature controlled power supply.
Substrate rotation with 0 to 25 rpm or better.

6. Sputter cathode specs
   6 number of cathodes for 2 inch dia targets.
   Manual cathode shutter
   Target to substrate distance 150 mm
   2 numbers of DC power supply 500 W or better

7. Evacuation pumping system
   Turbo molecular pump: 450 l/sec
   Roughing pump rotary: 350 l/ min

8. Gas inlet for Ar
   2 numbers Mass flow controllers Maximum of 50 sccm for Argon.
   1 numbers Mass flow controllers Maximum of 50 sccm for Nitrogen.
   1 numbers Mass flow controllers Maximum of 50 sccm for Oxygen.
9. System frame work 1 number
   Cooling water line for TMPs, cathodes and substrate heating system.
   Vacuum gages 2 numbers (from Atm pressure to UHV pressure)

10. Load lock chamber specs

    Material with SUS304 stainless steel only.

    With a dimension of ϕ 300 × 600

    ~ Ultimate pressure of 9 x 10^{-5} Pa.

    The chamber should have many kinds of flange ports as follows

    1 Port for Vacuum pump (Turbo molecular pump)

    1 Port for Vacuum gauge port of UHV

    1 Port for viewing

    Substrate holder replacement door flange

11. Substrate holder housing system –
    Three (3) numbers of storable substrate holders.
    Substrate is able to heat up to 600°C -1
    Substrate heating temperature is controlled by power supply -1

12. Reverse sputtering cleaning mechanism in load lock chamber
    RF power supply 500W with matching box.

13. RF sputter cathode specs in load lock
    Cathode of 2 inch dia 1 number.
    Manual cathode shutter
    Target to substrate distance 150 mm
    The power supply for RF cathode.
14. Substrate holder transfer with magnetic feedthrough type transfer rod manual operation
15. Vacuum gauge: 1 (from atmosphere to UHV pressure) for Load lock
16. Evacuation pumping system for load lock chamber
    Turbo molecular pump: 220l/sec
    Roughing pump (rotary pump): 250l/min
17. Control system and accessories.
18. Evacuation pumping control system
    Start-and-stop control of pump, open-and-close of valve by interlock, touch panel method controlled by sequencer.
19. Rake mounting for power supply and other control units.
    With one step down transfer (Available power 220 V 50Hz single phase and 420 V 50Hz three phase)
20. Baking system with RT-150 C max including sheath heater and temperature regulator with timer and power supply.
21. Cooling water supply unit. - Cooling capacity: about 2kW -Operating temperature limit:5-40℃ -Cooling method: Air cooling type -Refrigerant : R-407C
22. Consumable spare parts -Spare o-ring for chamber door , Cu gaskets for each flange, Spare shutter plates for sputter source and 1×Substrate heater bulb.
23. Guarantee items
    Thickness uniformity: $\pm$10% (50 ϕ dia at 2000 A° thickness)
    UHV pressure: Main sputtering chamber ~ $< 9 \times 10^{-8}$ Pa.
    Load lock chamber ~ $< 9 \times 10^{-5}$ Pa.

1. **Pre-Qualification criteria for the bidders:**

   a) The manufacturer of the Multitarget UHV sputtering chamber should have ISO-9001 quality management system to enforce standardized practices in design, quality control, manufacturing, testing and servicing of the equipment. The bidder/ supplier should have a proven record of manufacturing and supply of Multitarget UHV sputtering with at least 9×10^{-8} Pa base pressure.

   b) The bidder should preferably be the manufacturer of the chiller. In case, the bidder is not the original manufacturer then an authorization letter from the principal authorizing the bidder for sale and service of the chiller should be enclosed with the offer.

   c) The manufacturer should submit publications produced in the field of Spintronic devices like Magnetic tunnel junctions, gaint magnetoresistance devices, etc., using their Multitarget UHV system with at least 9×10^{-8} Pa base pressure. These should be submitted along with the technical bid.

2. **Electrical Works**

All the electrical and electronic should be compatible with 220 V and 50 Hz single phase and 420 V 50 Hz 3 phase power.

Information to be Provided with the Bid

a) Name of manufacturing unit, its address and contact details shall be declared.
b) Purchase order and relevant technical specification in support of "Qualification Criteria for Bidder".
c) Details of make, models and compliance standards of all the standard components with their catalogue copies.
d) Proposed overall dimensions, weight and mounting details.
e) Purchase order copies of the supplied chillers to Govt. of India labs or PSUs or stock listed industries must be attached with the bid. Bidder shall also submit complete address and contact details of the customers to whom they have supplied these chillers to verify the capability of the manufacturer to service and maintain them in satisfactory working condition.

3. **Warranty Period**:
The manufacturer/ supplier shall furnish a warranty against defective workmanship and materials for minimum period of 36 months from the date of commissioning for the full unit. He shall commit himself to speedy servicing, repair or replacement of defective parts of the complete unit during the period of said warranty.

UNIVERSITY OF HYDERABAD

(A Central University established by an Act of Parliament, 1974)

Prof. C. R. Rao Road, P.O. Central University Campus, Gachibowli,
Hyderabad - 500 046, Telangana, INDIA *website: www.uohyd.ac.in*

Department / Centre : Cryogenic Facility
Telephone: ... .

Email: **ammanabrolu@uohyd.ac.in**

Notice Inviting Tender No. …… Ref No: UoH/DST FIST/SS/ARK/01 … date: …15-4-2025

Sealed tenders are invited for supply of the following items under **single / two bid system** from the reputed manufacturers or their authorized dealers to the University of Hyderabad, Hyderabad.

Sealed bids should reach the office of the …Physics, Dr. A. Rajanikanth, School of Physics., University of Hyderabad, Hyderabad-500046, Telangana on or before -17 May - 2025 (date) 5:00PM (time).

The details of items, expected quantity and their technical specifications are given below:

S.N0.	Description of item and Specifications	Quantity (Unit)
1	**Description**	(Total Unit 1)
	1. Multi target UHV Sputtering System 2. The system should consist of deposition chamber and load lock chamber with Ultra high vacuum. 3. The sputtering chamber should be provided with 6 sputter cathodes of DC. 4. Load lock chamber with a rf sputtering gun for deposition of oxide materials. Specifications 4-1. Sputtering chamber specs Material with SUS304 stainless steel only.	

	With a dimension of ϕ 400 × 600 ~ Ultimate pressure of 9 x 10^{-8} Pa. The chamber should have many kinds of flange ports as follows 6 ports of r Sputter cathodes 1 Port for Vacuum pump (Turbo molecular pump with throttle butterfly valve) 1 Port for Vacuum gauge port of UHV 1 port for substrate holder with heating system 1 Viewing port 5. Substrate specifications Substrates size 2 inch diameter Substrate is able to heat up to 750° C or better Cathode shutters with manual operation Target to substrate distance is about 150 mm (Distance should be optimized with the magnetron design) Substrate heating with a temperature controlled power supply. Substrate rotation with 0 to 25 rpm or better. 6. Sputter cathode specs 6 number of cathodes for 2 inch dia targets. Manual cathode shutter Target to substrate distance 150 mm 2 numbers of DC power supply 500 W or better 7. Evacuation pumping system Turbo molecular pump: 450 l/sec Roughing pump rotary: 350 l/ min	

8. Gas inlet for Ar

   2 numbers Mass flow controllers Maximum of 50 sccm for Argon.

   1 numbers Mass flow controllers Maximum of 50 sccm for Nitrogen.

   1 numbers Mass flow controllers Maximum of 50 sccm for Oxygen.

9. System frame work 1 number

   Cooling water line for TMPs, cathodes and substrate heating system.

   Vacuum gages 2 numbers (from Atm pressure to UHV pressure)

10. Load lock chamber specs

    Material with SUS304 stainless steel only.

    With a dimension of ϕ 300 × 600

    ~ Ultimate pressure of 9 x 10^{-5} Pa.

    The chamber should have many kinds of flange ports as follows

    1 Port for Vacuum pump (Turbo molecular pump)

    1 Port for Vacuum gauge port of UHV

    Substrate holder replacement door flange

11. Substrate holder housing system –

    Three (3) numbers of storable substrate holders.

    Substrate is able to heat up to 600℃ -1

    Substrate heating temperature is controlled by power supply -1

12. Reverse sputtering cleaning mechanism

    RF power supply 500W with matching box.

13. RF sputter cathode specs

	Cathode of 2 inch dia 1 number. Manual cathode shutter Target to substrate distance 150 mm The power supply for RF cathode. 14. Substrate holder transfer with magnetic feedthrough type transfer rod manual operation 15. Vacuum gauge: 1 (from atmosphere to UHV pressure) 16. Evacuation pumping system for load lock chamber Turbo molecular pump: 220l/sec Roughing pump (rotary pump): 250l/min 17. Control system and accessories. 18. Evacuation pumping control system Start-and-stop control of pump, open-and-close of valve by interlock, touch panel method controlled by sequencer. 19. Rake mounting for power supply and other control units. With one step down transfer (Available power 220 V 50Hz single phase and 420 V 50Hz three phase) 20. Baking system with RT-150 C max including sheath heater and temperature regulator with timer and power supply. 21. Cooling water supply unit. - Cooling capacity: about 2kW - Operating temperature limit:5-40℃ -Cooling method: Air cooling type -Refrigerant ： R-407C 22. Consumable spare parts -Spare o-ring for chamber door , Cu gaskets for each flange, Spare shutter plates for sputter source and 1×Substrate heater bulb. 23. Guarantee items Thickness uniformity: $\pm$ 10% (50 ϕ dia at 2000 A° thickness) UHV pressure: Main sputtering chamber ~ $< 9 \times 10^{-8}$ Pa. Load lock chamber ~ $< 9 \times 10^{-5}$ Pa.	

Terms & conditions

S.	Clause	Description

No.		
1	Reference Number	The Reference No. … **UoH/DST FIST/SS/ARK/01** …. of the Notice Inviting Tender dated …------- … should be clearly mentioned on all envelopes and any correspondence including e-mails with the University.
2	Mode of submission	Hand-written (manuscript) bids and bids submitted through e-mail shall not be accepted. All pages of the tender document must be signed by the authorised representatives of bidder, in token of having accepted the terms and conditions incorporated in the tender notice.
3	Two Bid System *(This clause is applicable only when NIT is invited involving two bid system)*	As the purchase is a high value plant, machinery, etc., of a complex and technical nature, bids are required to be submitted in two parts as under: ***(i) Technical bid consisting of all technical details along with commercial terms and conditions; and*** ***(ii) Financial bid indicating item-wise price for the items mentioned in the technical bid.*** The technical bid and the financial bid should be sealed by the bidder in separate covers duly super-scribed and both these sealed covers are to be put in a bigger cover which should also be sealed and super-scribed. The technical bids are to be opened by the Purchasing Department / Centre at the first instance and evaluated by a duly constituted Purchase Committee. After such evaluation of technical bids, financial bids of only those firms/suppliers which are found technically suitable will be opened, for determining the lowest bid.
4	Certificate from original manufacturers	If the bid is not from the original equipment manufacturer, the suppliers/ dealers should submit an 'authorized agency certificate' from the manufacturer. Tenders received without Authorization Certificate from the original manufacturer will summarily be rejected. Imported purchases should normally be made directly from the foreign principal manufacturers. In the event of purchase made through an Indian agent of the foreign company, authorization letter issued by the foreign company / manufacturer should be submitted along with technical bid. No commission shall be payable to Indian Agent.
5	Make and Model	The Vendor/Supplier is required to clearly mention the make, model, size and other specifications along with the unit price. Items with better specifications may also be offered / quoted and the decision of the University in this regard is final. Appropriate brochures / pamphlets / literature (original and not photo copies) for each item providing full technical details must be attached to enable their verification.

6	Bidder Details	The bidder's complete correspondence address, phone numbers and e-mail ids, and website details (if any), must be clearly mentioned in the technical bid documents. (Pamphlet / Brochure depicting the profile and organisation structure may be enclosed.) The details of PAN, GSTIN, EPF, ESIC, Registered License Certificate, Labour License etc., may be mentioned and supporting documents shall be enclosed along with the technical bid.
7	Validity of bids	Prices quoted in the bid (financial bid) shall be valid for a minimum period of 90 days from the date of opening of the financial bids. No upward revision of the prices will be permitted during the said validity period.
8	Basic Price	The bidder should clearly indicate price per unit on FOR, UoH, Hyderabad basis. In respect of imported items, the prices should be quoted on the basis of CIP / CIF Hyderabad. *If the bidder quotes nil charges / consideration in respect of service contracts, the bid shall be treated as unresponsive and will not be considered.*
9	Taxes: GST& Customs Duty	The bidder should clearly indicate all applicable taxes separately for each item along with **HSN Code** and then arrive at a total price of all the items indicated in the Notice Inviting Tender. GST is applicable for all indigenous items. However, the University is entitled for concessional rate of GST @ 5% on all items mentioned vide GoI Notifications No. 45/2017-Central Tax (Rate) and 47/2017-Integrated Tax (Rate) dated 14.11.2017 read with GoI Notification No.09/2018 - Central Tax (Rate) dated 25.01.2018 as amended from time to time.
10	Other costs / charges	The bidder should clearly indicate all applicable other charges, if any, separately viz., transportation, packing, loading, unloading, insurance, installation, commissioning, testing, training, etc.,
11	Earnest Money Deposit (EMD) / Bid Security	The bidders should furnish bid security for Rs. _____/- (Rupees …… only) along with their technical bid in the form of Account Payee Demand Draft / Banker's Cheque from any of the Commercial Banks in an acceptable form, drawn in favour of Finance Officer, University of Hyderabad, payable at Hyderabad. If the bidder furnishes a Fixed Deposit Receipt / Bank Guarantee, the instrument shall be issued in favour of Finance Officer, University of Hyderabad, payable at Hyderabad and shall be valid for a period of 135 days from the date of opening technical bids.

		The MSE units registered with MSME and certificates obtained from NSIC under the Single Point Registration Scheme (SPRS) shall be exempted from payment of Earnest Money Deposit (EMD) on production of requisite proof in respect of valid certification from NSIC.
12	Warranty	The bidders should offer a minimum period of 3 years warranty for all the quoted items from the date of successful installation of the equipment/ item. The quoted price shall be inclusive of warranty costs. AMC/FMC charges for fourth and fifth year shall be quoted separately, if applicable.
13	Comprehensive Annual Maintenance Contract (CAMC)	Bidders shall quote Comprehensive Annual Maintenance (CAMC) charges for a period of 2 years after expiry of the initial warranty period of 3 year. In all other cases, rate for at least three years' CAMC be quoted.
14	Agreement	An Agreement in the format prescribed by the University of Hyderabad is required to be entered into by the Suppliers or their authorised Representative with the University in respect of purchase of Goods / Services valuing Rs. 10 lakh and above, within a period of 15 days from the date of issue of Purchase Order / Work Order.
15	Performance Security / Performance Bank Guarantee	Successful bidder should submit a Performance Security (in all cases of purchases valuing above Rs. 10 lakh in the form of Performance Bank Guarantee for an amount equivalent to 3% of the total basic price of the Goods / Services (exclusive of taxes and duties) within a period of 15 days from the date of issue of Purchase Order/ Work Order. Performance Security should remain valid for the entire warranty period plus additional (grace) period of 60 days, from the date of successful installation of the equipment/item. Performance Security shall be submitted in the form of Account Payee Demand Draft / Fixed Deposit Receipt / Banker's Cheque / Bank Guarantee from any of the Commercial Banks in an acceptable form, drawn in favour of Finance Officer, University of Hyderabad, payable at Hyderabad.
16	Payment	**97 %** of the total cost will be released after delivery, satisfactory installation, commissioning, testing and training (if any), against submission of valid TAX Invoice (with **HSN** code) with GSTIN number of the supplier. The remaining **3 %** will be released on submission of a Performance Bank Guarantee (PBG) for an equivalent amount. In case of purchases from foreign manufacturers/suppliers, Letter of Credit (LC) will be opened for 100% of the purchase value and **92.5 %** shall be released after signing the Agreement and shipment / submission

		of documents. As regards balance payment, it will be released on submission of PBG for equivalent amount (in Indian currency) or completion of the warranty period as per the terms included in the NIT and Purchase Order.
17	Delivery Location	The items shall be delivered to: Dr. A. Rajanikanth Department / Centre: ……Physics…….. School: ……Physics……... University of Hyderabad, Gachibowli, Hyderabad - 500 046, India
18	Delivery Period	The supplier should be able to deliver the items within …180……. days *(mention the required delivery period by the proposer)* from the date of issue of the purchase order.
19	Liquidated damages for late Delivery	In the event of delay in delivery beyond stipulated period indicated in the Purchase Order, liquidated damages @ 1% per each week of delay subject to a maximum of 5% of the cost of the order will be levied and collected by the University, by way of deduction from the payments due to the vendor/supplier.
20	Risk Purchase Clause	If the Supplier fails to deliver the ordered materials within the stipulated delivery period specified in the Purchase Order, the University may resort to procure such items and in such a manner as deemed appropriate, goods or services similar to those undelivered, and the Supplier will be liable to reimburse the University for any excess costs for such similar goods or services.
21	Right to reject	The University reserves the right to reject any or all the bids or accept a part of the quotation without assigning any reasons therefor. The decision of the University in this regard is final and binding on all the suppliers.
22	Resolution of Disputes	Any disputes arising out of this contract shall be referred to the University, and if any of the parties hereto is dissatisfied with the decision, the dispute shall be referred to the decision of the Arbitrator, who should be acceptable to both the parties, to be appointed by the Vice-Chancellor of the University. The decision of such Arbitrator shall be final and binding on both the parties.
23	Jurisdiction of Courts	Disputes, if any, arising during course of execution of the order are to be settled within the jurisdiction of Hyderabad / Ranga Reddy District Courts only.
24	Opening of tenders / bids	The bids will be opened in the office of ……**IAO….,** University of Hyderabad, Hyderabad on **-19-05-2025-**... (date) at -3:30- PM…….. (time) in the presence of bidders or their authorised representatives.

		The authorised representative shall bring the authorisation letter and submit the same to the University before opening of bids. On failure to do so, the Representative loses the right to participate in the bid opening process.
25	Delay in submission	The University will not in any way be responsible for any postal / courier delay. Bids received beyond the stipulated date and time of submission are summarily rejected. Tenders incomplete in any respect will be summarily rejected.
26	Due date of submission of bids	Sealed Bids should be submitted to the office of …Prof. S. Srinath, School of Physics, University of Hyderabad, Gachibowli, Hyderabad - 500 046 on or before --…. (date) -- PM (time). Bids received thereafter shall not be considered. *(Minimum time to be allowed for submission of bids should be three weeks for procurement of indigenous items and four weeks in case of imported items from the date of publication of the Notice Inviting Tender in CPP Portal and University of Hyderabad Website)*

Signature of Dean/Head/Director/PI/Faculty In-charge
Name: ……Prof. S. Srinath…….……
Dr. A. Rajanikanth
Department / Centre: Cryogenic Facility
Phone No. Ph: (040)23134330(M); 9440491259 (M); 8331879303
E-mail ID: srinath@uohyd.ac.in; srinaths10@gmail.com